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Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

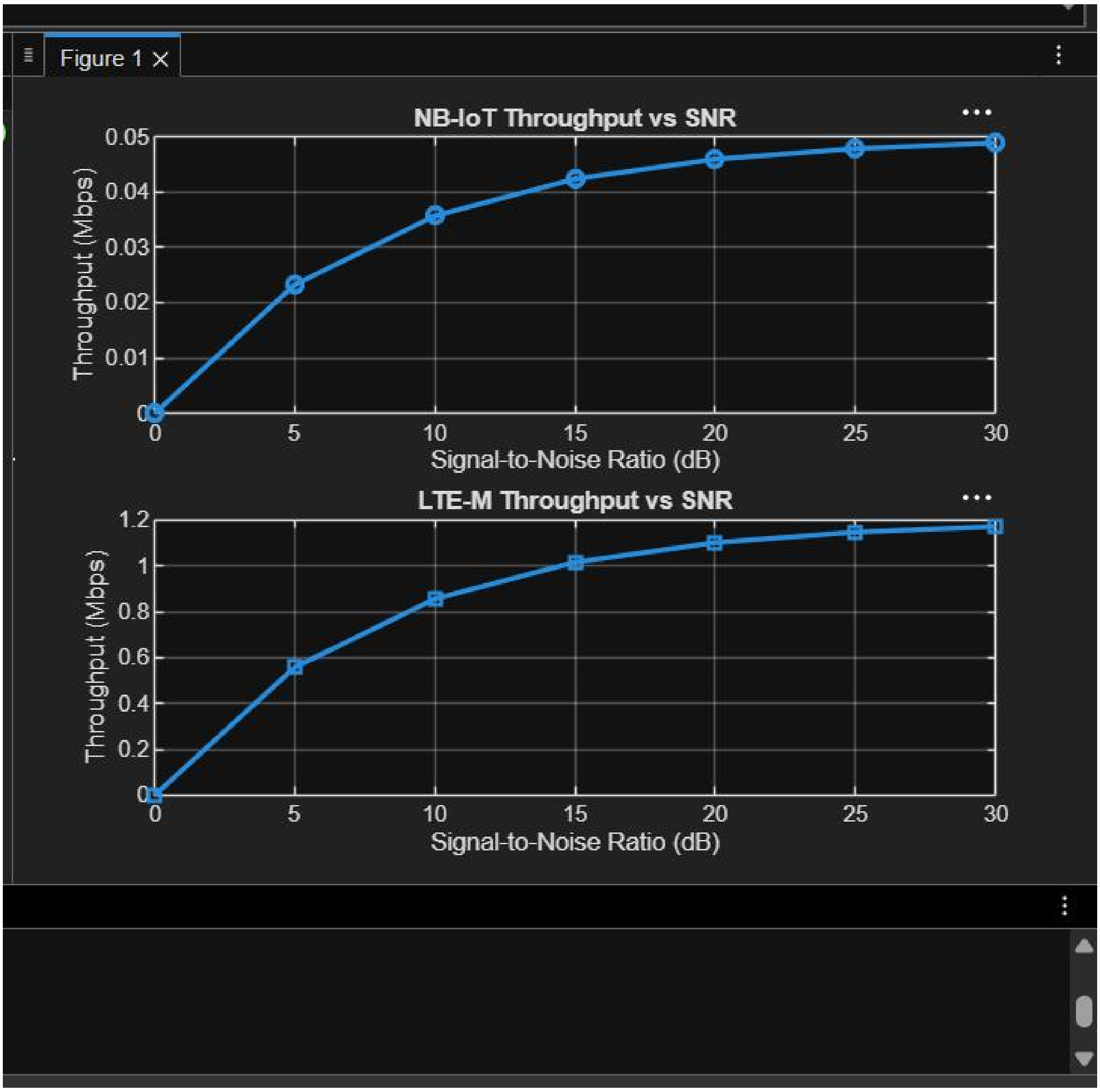
OBJECTIVE :

CODE:

```
clc;
clear;
close all;

% SNR values (dB)
SNR = 0:5:30;

% Simulated Throughput (Mbps)
NB_IoT = 0.05 * (1 - exp(-SNR/8));
LTE_M = 1.2 * (1 - exp(-SNR/8));

% Display values
disp('SNR (dB)    NB-IoT (Mbps)    LTE-M (Mbps)')
disp([SNR' NB_IoT' LTE_M'])

figure

% -------- Graph 1 --------
```

```
subplot(2,1,1)
plot(SNR, NB_IoT, '-o', 'LineWidth', 2)
title('NB-IoT Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
grid on

% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR, LTE_M, '-s', 'LineWidth', 2)
title('LTE-M Throughput vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Throughput (Mbps)')
```

grid
on



OBJECTIVE 2:

CODE:

clc;

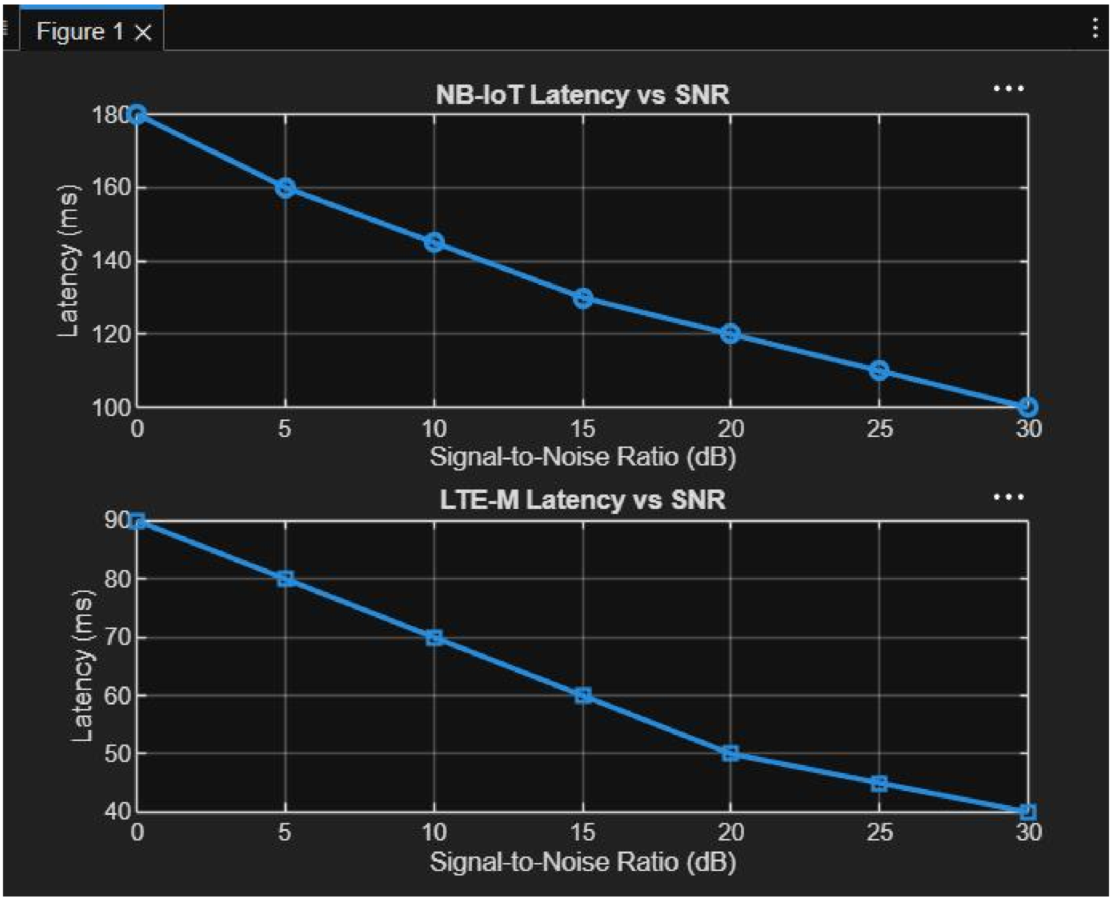
clear;

close all;

% SNR values (dB)

```
SNR = 0:5:30;

% Simulated Latency (ms)
NB_IoT = [180 160 145 130 120 110 100];
LTE_M = [90 80 70 60 50 45 40];

% Display values
disp('SNR(dB)   NB-IoT Latency(ms)   LTE-M Latency(ms)')
disp([SNR' NB_IoT' LTE_M'])

figure

% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR, NB_IoT, '-o', 'LineWidth', 2)
title('NB-IoT Latency vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on

% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR, LTE_M, '-s', 'LineWidth', 2)
title('LTE-M Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Latency (ms)')
grid on
```



OBJECTIVE 3:

CODE:

```
clc;
clear;
close all;
% SNR values (dB)
SNR = 0:5:30;
% Simulated Packet Delivery Ratio (%)
```

```
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])

figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(SNR,NB_IoT,'-o','LineWidth',2)
title('NB-IoT Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
% -------- Graph 2 --------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```

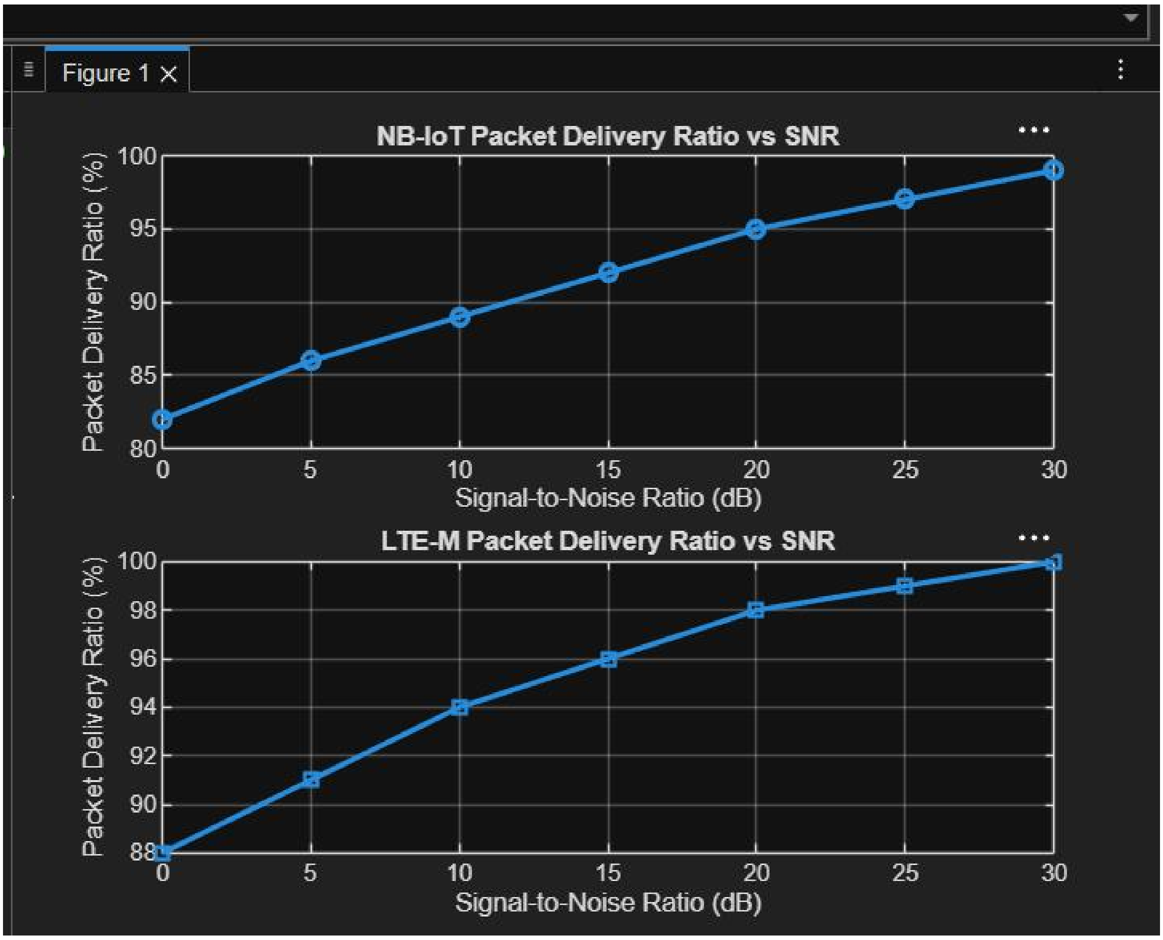


Figure 1
NB-IoT Packet Delivery Ratio vs SNR
Packet Delivery Ratio (%)
100
95
90
85
80
0
5
10
15
20
25
30
Signal-to-Noise Ratio (dB)
LTE-M Packet Delivery Ratio vs SNR
Packet Delivery Ratio (%)
100
98
96
94
92
90
88
0
5
10
15
20
25
30
Signal-to-Noise Ratio (dB)